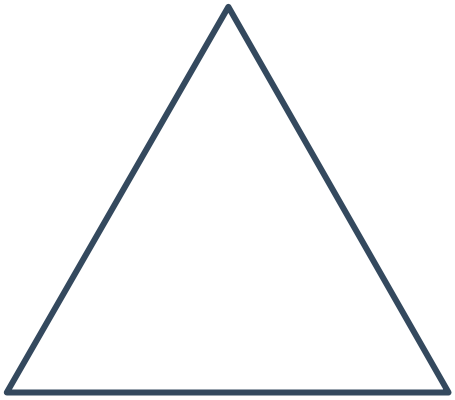
# Symmetry

**1** Consider all the single digit numbers shown below.
Which of these numbers have line symmetry?

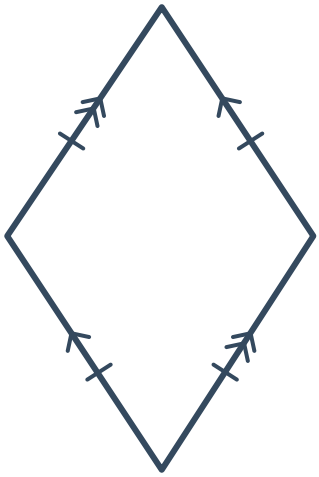
**1 2 3 4 5 6 7 8 9 0**

**2** How many lines of symmetry do the following shapes have?

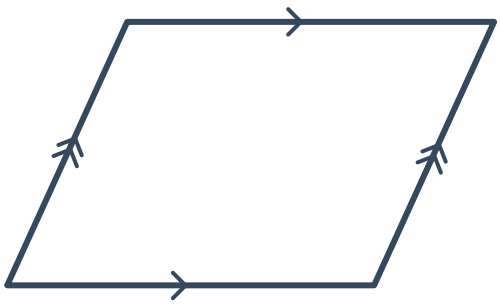
**a**



**b**



**c**

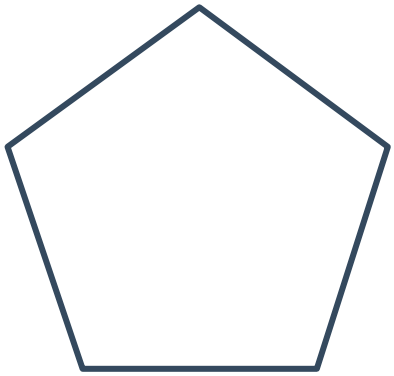


**d**

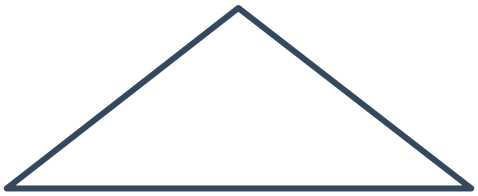


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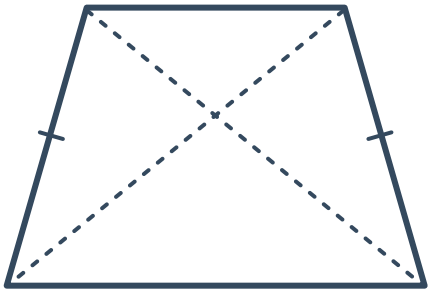
e



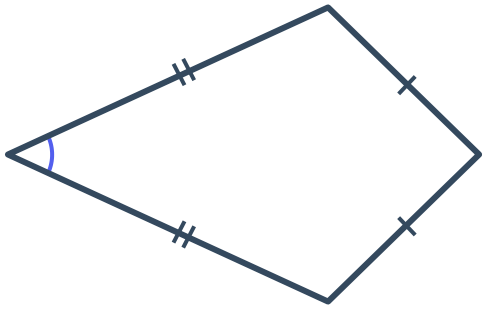
f



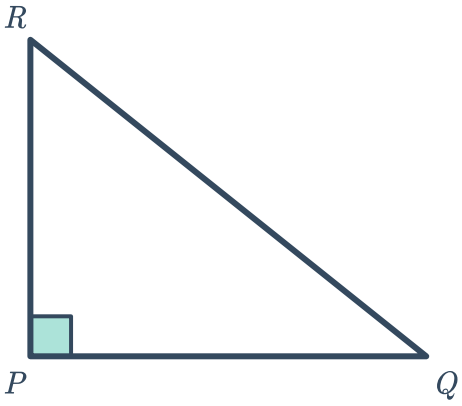
g



h

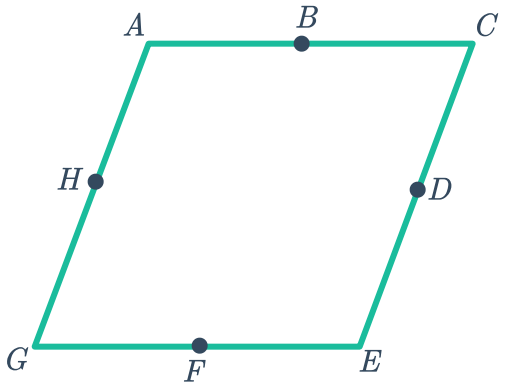


3 Consider the $\triangle RPQ$ shown. The triangle is reflected across $\overline{PQ}$.
Identify the shape formed by the triangle and its reflection.

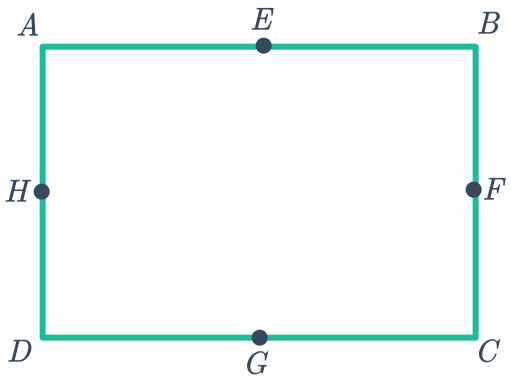



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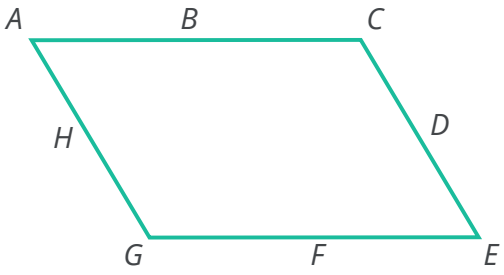
**4** Consider the following rhombus.
The points $B, D, F$ and $H$ represent the midpoints of the sides of the rhombus.
Write down the pairs of points that can be joined to form a line of symmetry for the rhombus.




**5** Consider the following rectangle.
The points $F, G, H$ and $E$ represent the midpoints of the sides of the rectangle.
Write down the pairs of points that can be joined to form a line of symmetry for the rectangle.

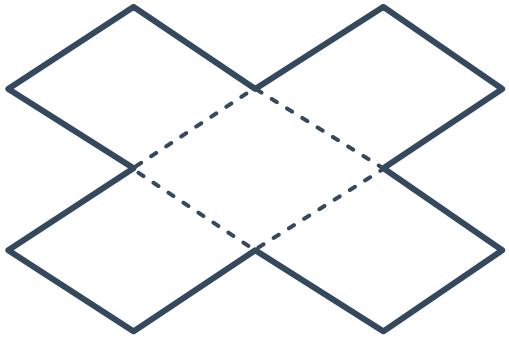



**6** Consider the following parallelogram.
The points $B, F, H$ and $d$ represent the midpoints of the sides of the parallelogram.
Write down the pairs of points that can be joined to form a line of symmetry for the parallelogram.



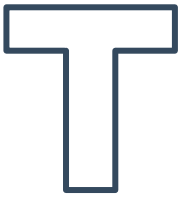

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**7** The following shape is constructed from five identical rhombuses.
How many lines of symmetry does this shape have?



**8** Determine whether the following diagrams have rotational symmetry.

**a**



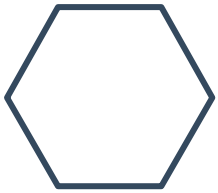
**b**



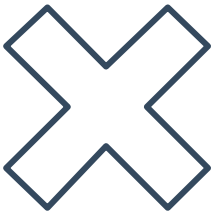
**c**



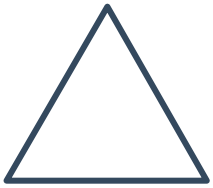
**d**



**e**



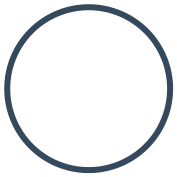
**f**



**g**



**h**



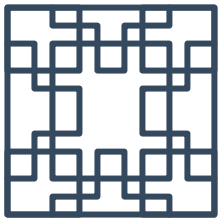
**i**



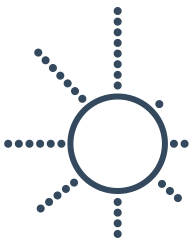
**j**



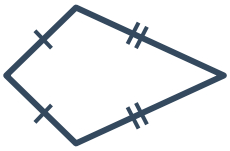
**k**



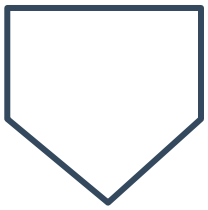
**l**



**m**



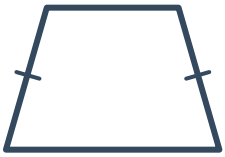
**n**



**o**

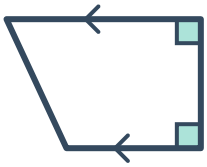


**p**

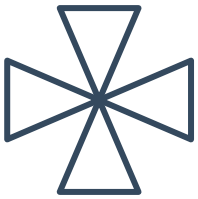


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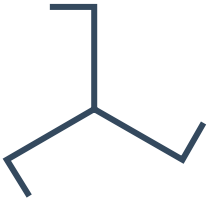
q



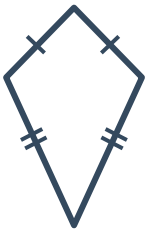
r



s



t



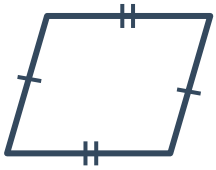
u



v



w



**9** Determine whether the following letters have rotational symmetry.

a



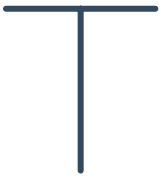
b



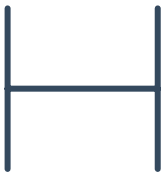
c



d



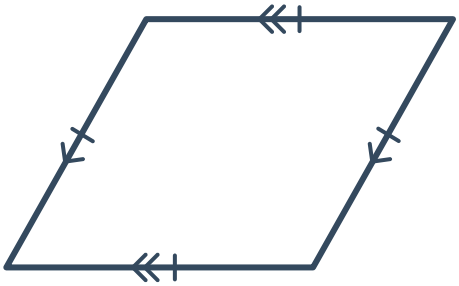
e



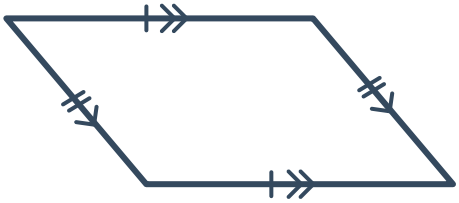
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**10** State the order of rotational symmetry for the following shapes.

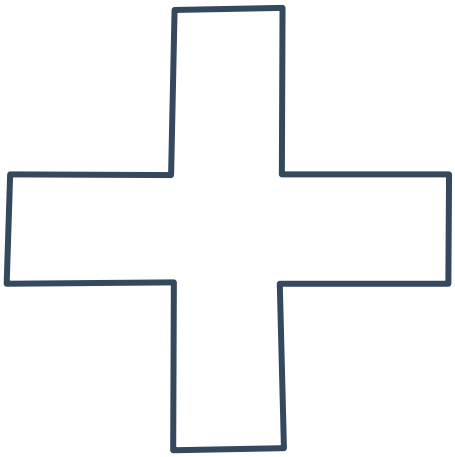
**a**



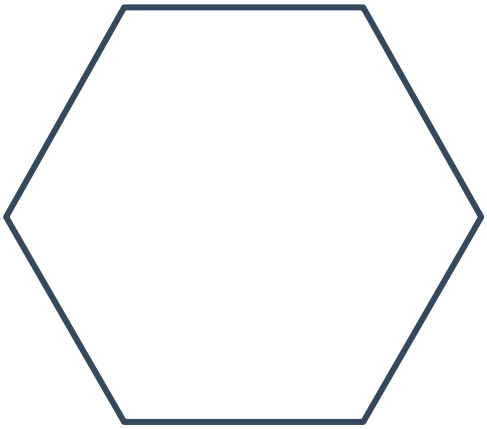
**b**



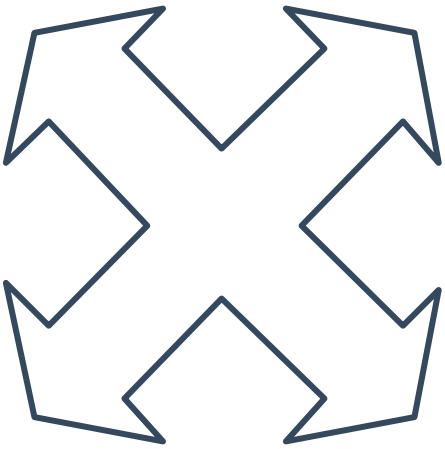
**c**



**d**



**e**

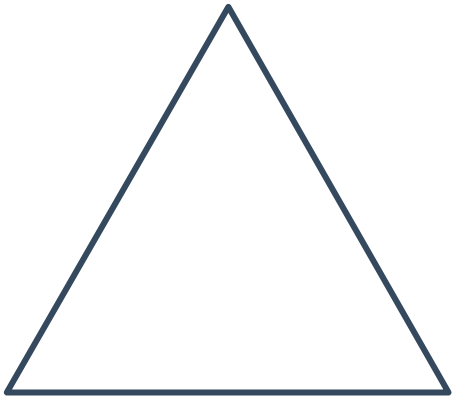


**f**

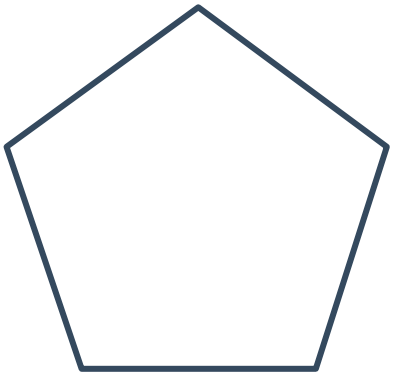


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g



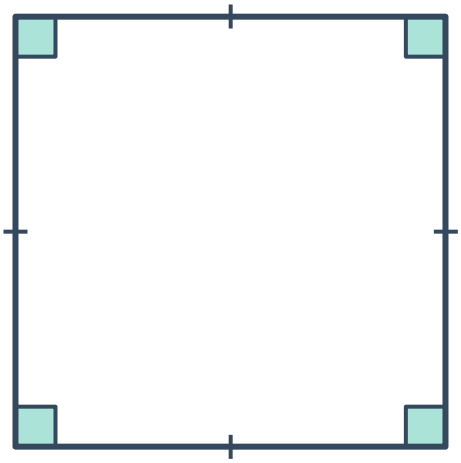
h



i



j

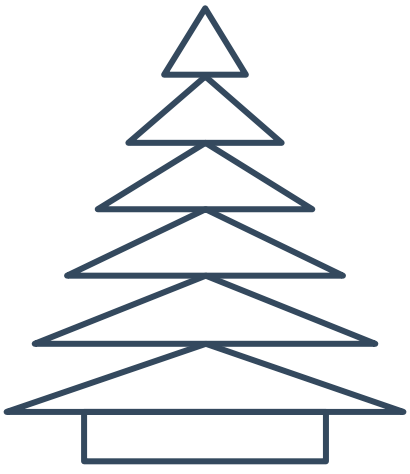


11 State the type(s) of symmetry each of the following diagrams have.

a

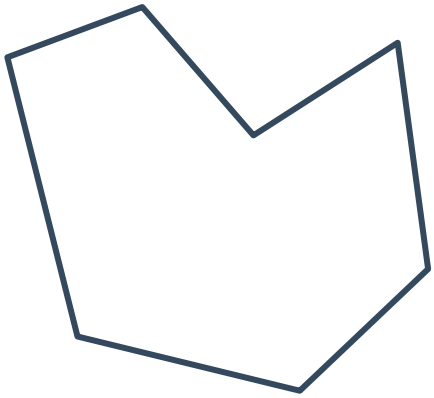


b

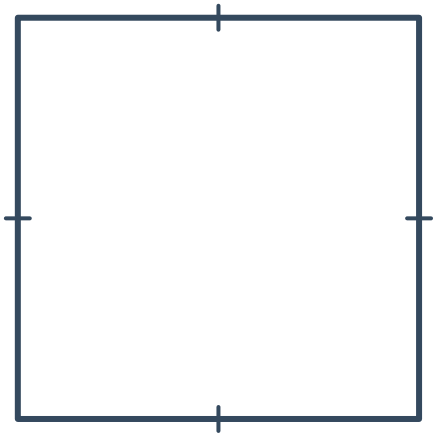


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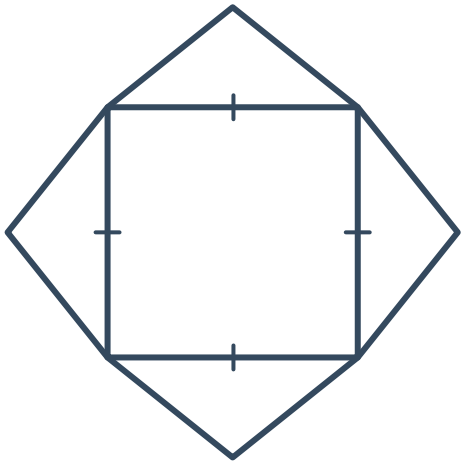
c



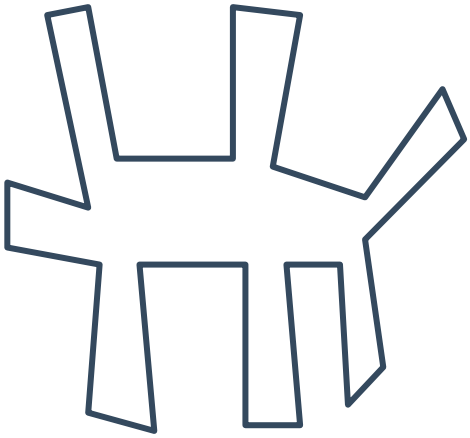
d



e



f



**12** If a shape has a rotational symmetry of order 2, state the size of each angle of rotation.

**13** Do all isosceles triangles have rotational symmetry?

**14** Do the following road signs have rotational symmetry?

a



b




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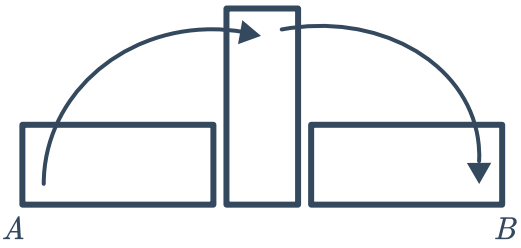
**15** Consider the road sign shown.



**a** State the order of rotational symmetry.

**b** Does the road sign have line symmetry?

**16** Dave wants to get a heavy rectangular box from point $A$ to point $B$ by repeatedly turning it over. The width of the box is $30$ cm and its length is twice its width.



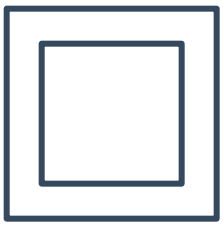

**a** The diagram shows the distance travelled after the first and second turn. After $3$ turns, what distance along the ground has been covered starting from point $A$?

**b** If the distance from point $A$ to point $B$ is $1620$ cm, how many times must the box be turned?

## Additional questions

**17** Determine whether the following shapes have rotational symmetry:

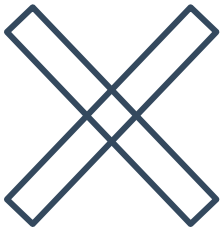
**a**



**b**



**c**



**d**



**18** If a shape has a rotational symmetry of order $4$, state the size of each angle of rotation.

**19** Do all equilateral triangles have rotational symmetry?

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